Contribution Journal: Lab 06 Regression and Classification

Name: Joseph Clay
Group name: N/A (individual submission)
Group members: Joseph Clay
Course: ITAI 1371
Date: October 14, 2026

1. My contributions

Date	What I did	Where it is (file, cell, commit)
Sun Oct 11	Set up the project: Python virtual environment, Git repository, `.gitignore`, and a working copy of the starter notebook.	Commit `cf3a993` (6:34 PM)
Sun Oct 11	Completed Task 1: built, trained, and evaluated the Linear Regression model for `Fare`. Result: MSE 3364.92, RMSE $58.01. Consolidated my work into one notebook named after the Canvas original.	Task 1 code cell; commit `9b126e0` (8:11 PM)
Mon Oct 12	Completed Task 2: built, trained, and evaluated the Logistic Regression model for `Survived`. Result: 74.83% accuracy. Wrote Knowledge Check answers 1 to 3 and printed the model coefficients. Ran Restart & Run All with no errors.	Task 2 code cell, coefficient cell, Knowledge Check markdown cells; commit `b245800` (4:18 PM)
Tue Oct 13	Exported the fully executed notebook with all outputs to PDF.	`L06_JosephClay_ITAI1371.pdf`
Wed Oct 14	Prepared the reflective and contribution journals and organized the final deliverables.	`journals/`, journal PDFs

2. Code I wrote or ran myself

- **Task 1 cell (Linear Regression):** created the `LinearRegression()` model, trained it with `fit` on the training set, predicted on the test set, and scored it with `mean_squared_error`. I also imported NumPy and added a line that prints the square root of the MSE. That turns the error back into money (58.01, which the notebook labels as dollars, though the Titanic fares are actually 1912 British pounds). That made the result easier to understand than the squared value of 3364.92.

- **Task 2 cell (Logistic Regression):** created the `LogisticRegression()` model, trained it, predicted survival on the test set, and scored it with `accuracy_score` (74.83%).
- **Coefficient cell:** added a code cell that prints `lr_model.coef_` and `lr_model.intercept_` for Knowledge Check 2. Output: Age −0.437, Pclass −38.297, intercept 133.31.
- **Clean run:** ran Restart & Run All so the cells execute in order (execution counts 1 to 5) with no errors, which confirms that every output in the PDF comes from a fresh run.

3. Documentation I wrote

- **Knowledge Check 1:** my explanation of regression vs. classification. Regression answers "how much or how many," and classification sorts things into buckets ("which one? yes or no?").
- **Knowledge Check 2:** my interpretation of the coefficients. Each extra year of age lowers the predicted fare by about $0.44. Each step down in class (1st to 2nd to 3rd) lowers it by about $38.30. The intercept is the baseline where the line starts.
- **Knowledge Check 3:** why MSE fits regression (it measures distance from a continuous value) and why accuracy fits classification (it counts right vs. wrong), so accuracy does not work for fares.
- Step-by-step comments in both task cells (create, train, predict, evaluate).
- Commit messages that record the result of each stage.

4. Team members' contributions

I worked alone on this lab. All of the notebook code, the Knowledge Check answers, and the deliverables are my individual work. I studied with the lab kit's brief and notebook guide. I used Claude Code, an AI coding assistant, only for mechanical tasks: setting up the virtual environment and repository, exporting the PDFs, and drafting this contribution log from my commit history.

5. GitHub evidence

- **Repository:** `https://github.com/ClayClimate-AI/itai-1371-lab06-regression-classification`
- `cf3a993`, Oct 11, 6:34 PM: Initial commit: Lab 06 kit scaffolding, working notebook copy, and .gitignore
- `9b126e0`, Oct 11, 8:11 PM: Task 1: Linear Regression complete (MSE 3364.92, RMSE $58.01)
- `b245800`, Oct 12, 4:18 PM: Complete notebook: Task 1, Task 2 (74.83%), Knowledge Check Q1–Q3; clean Restart & Run All

6. What I would do differently as a teammate

I finished the notebook early, but I left the PDF export and the journals until the last two days. Next time I will write the journals while the work is still fresh, and I will commit and push each

deliverable as soon as it is done, so the repository always shows where the work stands.